

- Q.28 Describe the automated electrical conductivity method of blood cell counting.
- Q.29 Write applications of centrifuge in blood sample analysis.
- Q.30 Explain working principle and use of incubator.
- Q.31 Write a short note on autoclave and its applications.
- Q.32 Explain principle and applications of chromatography in clinical labs.
- Q.33 Describe principle and applications of electrophoresis.
- Q.34 Explain types of microscopes with their applications.
- Q.35 Write application of auto-analyzer in biochemistry labs.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 With neat diagram, explain the working of a photo colorimeter.
- Q.37 Explain the working principle and block diagram of pH meter.
- Q.38 Describe the principle, construction & application of blood cell counters.

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Roll No.

4th Semester/ Med. Eltx.

**Subject : Medical Laboratory Instruments/
Analy Inst. (BM)**

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The light source commonly used in a spectrophotometer is:
- a) Tungsten lamp b) LED
- c) Sodium lamp d) Mercury ARC lamp
- Q.2 A photo colorimeter measures:
- a) Blood Cell count
- b) Absorbance of light by solution
- c) pH of blood
- d) Air pressure
- Q.3 Which component of flame photometer supplies compressed air?
- a) Atomizer b) Air Compressor
- c) Optical filter d) Detector
- Q.4 The electrode used in pH meter is :
- a) Calomel electrode b) Glass electrode
- c) Copper electrode d) Carbon electrode

- Q.5 Which section of auto-analyzer records the final result?
- a) Mixer b) Sampler
c) Recorder d) Dialyzer
- Q.6 The automated optical method of blood cell counting uses:
- a) Light scattering b) Conductivity
c) Colorimetry d) Chromatography
- Q.7 The microscope that uses visible light and lenses is :
- a) Compound microscope
b) Electron microscope
c) Phase contrast microscope
d) Scanning microscope
- Q.8 Ovens in laboratories are mainly used for:
- a) Cooling b) Heating and drying
c) Sterilizing with steam d) Blood analysis
- Q.9 In chromatography, the stationary phase is usually a:
- a) Liquid solvent
b) Solid or liquid on solid support
c) Gas
d) Flame
- Q.10 Electrophoresis separates molecules based on :
- a) Size & electrical charge
b) Color & brightness
c) Density
d) Absorption

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Write one use of spectrophotometer in biochemistry.
- Q.12 Name any one type of monochromator.
- Q.13 Write the principle of flame photometer in one line.
- Q.14 Mention one application of pH meter.
- Q.15 Name two sections of an auto-analyzer.
- Q.16 Expand DNA.
- Q.17 Write one use of centrifuge in pathology labs.
- Q.18 Name the sterilization method used for surgical instruments.
- Q.19 Write one type of microscope.
- Q.20 Mention one application of chromatography.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the principle of a photo colorimeter.
- Q.22 Write a short note on light sources used in spectrophotometers.
- Q.23 Explain the construction and working of a pH meter.
- Q.24 Explain the importance of blood pH measurement.
- Q.25 Write applications of flame photometer in clinical chemistry.
- Q.26 Describe the components of a flame photometer with neat block diagram.
- Q.27 Explain the working principle of a blood cell counter.